

Missing Data and Low-Oxygen Exposure in Louisiana Coastal Monitoring

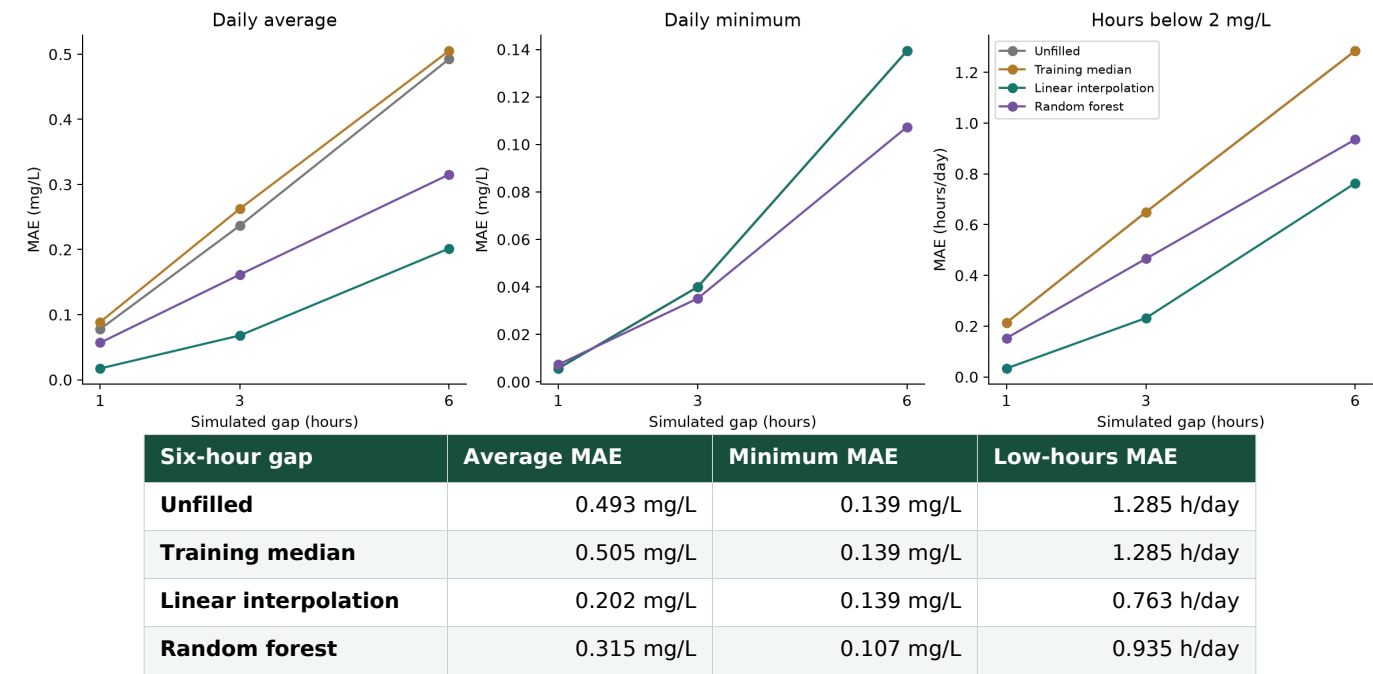
Independent methods case study using public hourly USGS observations from Wilkinson Bayou, Louisiana

Research question

When consecutive oxygen readings are missing, how well do common methods preserve the daily average, daily minimum, and hours below an exploratory 2 mg/L threshold?

Data and design

The public record contains 9,069 rows from June 2022 to October 2023, including 8,440 oxygen observations. Training precedes July 1, 2023; testing uses 113 complete days in July-October. Identical interior 1-, 3-, and 6-hour gaps retain observed endpoints. Each duration is repeated 20 times per day. Methods are unfilled data, a training median, linear interpolation, and a random forest.



Finding

In this experiment, interpolation best preserved average oxygen and approximate low-oxygen duration. The random forest produced the smallest daily-minimum error and missed fewer low-oxygen days during six-hour gaps, but it was worse on the other summaries. Every method undercounted low-oxygen hours on average. Method choice therefore depends on the research summary being protected.

Practical value

The workflow helps researchers decide whether incomplete days can support a specific analysis and how reconstructed values should be labeled. Applied to approved Lake Maurepas records, it could test the team's actual sensor frequency, outage patterns, and biologically relevant summaries.

Scope

This coastal salt-marsh station is not Lake Maurepas. Low hours count hourly samples below 2 mg/L, not continuous exposure. Unfilled gaps count only observed low readings. Repeated masks are dependent; simulated interior gaps may differ from real failures. Interpolation is retrospective, the model retains other sensor records (existing gaps use training medians), and 2 mg/L is an exploratory threshold rather than a site-specific standard. Results do not establish ecological impact, causation, regulatory compliance, or a safe gap length.

Source: Mize et al. (2025), USGS data release 10.5066/P13GBADQ. Analysis code, trial results, and full methods are included in the project package.